# Nathan J. Szymanski

Assistant Professor · Materials Science and Engineering

*410 Westwood Plaza, Los Angeles, CA 90095*

nszymanski@seas.ucla.edu | szymanski-research-group.com

## Research Interests

I am a materials scientist working to advance the development of sustainable energy technologies by integrating computational and experimental methods. My work combines *ab-initio* quantum chemistry, machine learning, and Monte Carlo simulations to model inorganic materials synthesis. By pairing these techniques with *in-situ* characterization and AI-driven data analysis, my group streamlines the process between initial discovery and commercialization of next-generation materials for energy and high-temperature applications.

## Education

**University of California, Berkeley** — *Berkeley, CA*
PhD Materials Science and Engineering — *2019 - 2024*

**University of Toledo** — *Toledo, OH*
BS Physics and Applied Mathematics — *2015 - 2019*

## Professional Experience

| | |
|---|---|
| 2025- | **Assistant Professor**, Materials Science and Engineering, University of California, Los Angeles |
| 2024-2025 | **Postdoctoral Researcher**, Chemical Engineering and Materials Science, University of Minnesota |
| 2019-2024 | **Graduate Research Assistant**, Materials Science and Engineering, University of California, Berkeley |
| 2023-2024 | **Graduate Teaching Assistant**, Graduate-level Thermodynamics, University of California, Berkeley |

## Publications

1. X. Tong, E. Jin, J. Xu, A. Rao, P. Jiang, and **N. J. Szymanski**, Scalable machine learning framework for multi-phase identification from powder X-ray diffraction. *arXiv:2609.06908* (2026).

2. A. Giunto, Y. Fei, P. Nevatia, B. Rendy, **N. J. Szymanski**, and G. Ceder, Accurate SEM-EDS quantification, automation, and machine learning enable high-throughput compositional characterization of powders. *Nature Communications* (2026).

3. **N. J. Szymanski**, K. J. Warren, A. W. Weimer, and C. J. Bartel, Cation Vacancies Mediate Thermochemical Water Splitting with Iron Aluminates. *Chemistry of Materials* 38, 3145–3156 (2026).

4. M. Abolhasani, M. Ahmadi, S. Baird, C. P. Berlinguette, K. A. Brown, D. Fenning, I. T. Foster, W. Gottstein, S. Pablo-Garcia, G. Gomes, J. R. Hattrick-Simpers, J. E. Hein, T. Hitosugi, S. V. Kalinin, B. Maruyama, J. Schrier, M. Seifrid, T. D. Sparks, S. Sun, B. R. Sutherland, **N. J. Szymanski**, D. Thomas, L. Walters, J. Xu, Y. Zeng, C. Yang, On the Need for Autonomous Science Instruments: A Call to Action. *ChemRxiv* (2026).

5. J. Schlesinger, S. Hjaltason, **N. J. Szymanski**, and C. J. Bartel, Thermodynamic assessment of machine learning models for solid-state synthesis prediction. *Materials Horizons* (2026).

6. A. J. Lannerd, **N. J. Szymanski**, and C. J. Bartel, Thermodynamics of proton insertion across the perovskite–brownmillerite transition in $La_{0.5}Sr_{0.5}CoO_{3-\delta}$. *Phys. Rev. Materials* 10, 025401 (2026).

7. L. Rettenberger, **N. J. Szymanski**, A. Giunto, O. Dartsi, A. Jain, G. Ceder, V. Hagenmeyer, and M. Reischl, Leveraging unlabeled SEM datasets with self-supervised learning for enhanced particle segmentation. *npj Computational Materials* 11, 289 (2025).

8. F. Liu, **N. J. Szymanski**, K. Noordhoek, H. Shin, C. J. Bartel, and B. Jalan, Unraveling the Growth Dynamics of Rutile $Sn_{1-x}Ge_xO_2$ Using Theory and Experiment. *Nano Letters* 25, 299–305 (2025).

9. **N. J. Szymanski** and C. J. Bartel, Establishing baselines for generative discovery of inorganic crystals. *Materials Horizons* 12, 8000–8011 (2025).

10. **N. J. Szymanski**, A. Smith, P. Daoutidis, and C. J. Bartel, Topological descriptors for the electron density of inorganic solids. *ACS Materials Letters* 7, 2158–2164 (2025).

11. S. Sullivan, S. Lee, **N. J. Szymanski**, A. Merchant, E. D. Cubuk, T. Low, and C. J. Bartel, Computational search for materials having a giant anomalous Hall effect in the pyrochlore and spinel crystal structures. *Phys. Rev. Materials* 9, 114409 (2025).

12. H. Kim, K. Jun, **N. J. Szymanski**, V. Sai, Z. Cai, M. Crafton, G.-H. Lee, S. E. Trask, F. Babbe, Y.-W. Byeon, P. Zhong, D. Lee, B. Park, W. Jung, B. D. McCloskey, and W. Yang, Screening and Development of Sacrificial Cathode Additives for Lithium-Ion Batteries, *Advanced Energy Materials* 15, 2403946 (2025).

13. V. S. Avvaru, T. Li, G.-H. Lee, Y.-W. Byeon, K. P. Koirala, O. J. Marques, B. L. D. Rinkel, Y. Fu, D. Milsted, S. Jeong, **N. J. Szymanski**, M. Kunz, F. Babbe, E. Lee, V. Battaglia, B. D. McCloskey, J. N. Weker, C. Wang, W. Yang, R. J. Clément, and H. Kim, Alternative Solid-State Synthesis Route for Highly Fluorinated Disordered Rock-Salt Cathode Materials for High-Energy Lithium-Ion Batteries. *Advanced Energy Materials* 15, 2500492 (2025).

14. Y. Chen, X. Zhao, K. Chen, K. P. Koirala, R. Giovine, X. Yang, S. Wang, **N. J. Szymanski**, S. Xiong, Z. Lun, H. Ji, C. Wang, J. Bai, F. Wang, B. Ouyang, G. Ceder, Coherent-Precipitation-Stabilized Phase Formation in Over-Stoichiometric Rocksalt-Type Li Superionic Conductors, *Advanced Materials* 37, 2416342 (2025).

15. **N. J. Szymanski**, Y.-W. Byeon, Y. Sun, Y. Zeng, J. Bai, M. Kunz, D.-M. Kim, B. A. Helms, C. J. Bartel, H. Kim, and G. Ceder, Quantifying the regime of thermodynamic control for solid-state reactions during ternary metal oxide synthesis. *Science Advances* 10, eadp3309 (2024).

16. **N. J. Szymanski** and C. J. Bartel, Computationally Guided Synthesis of Battery Materials. *ACS Energy Letters* 9, 2902–2911 (2024).

17. Y. Fei, B. Rendy, R. Kumar, O. Dartsi, H. P. Sahasrabuddhe, M. J. McDermott, Z. Wang, **N. J. Szymanski**, L. N. Walters, D. Milsted, Y. Zeng, A. Jain, G. Ceder, AlabOS: a Python-based reconfigurable workflow management framework for autonomous laboratories. *Digital Discovery* 3, 2275–2288 (2024).

18. S. Wang, **N. J. Szymanski**, Y. Fei, W. Dong, J. N. Christensen, Y. Zeng, M. Whittacker, and G. Ceder, Direct Lithium Extraction from $\alpha$-Spodumene through Solid-State Reactions for Sustainable $Li_2CO_3$ Production. *Inorganic Chemistry* 63, 13576–13584 (2024).

19. **N. J. Szymanski**, S. Fu, E. Persson, and G. Ceder, Integrated analysis of X-ray diffraction patterns and pair distribution functions for machine-learned phase identification. *npj Computational Materials* 10, 45 (2024).

20. Y. Zeng* and **N. J. Szymanski***, T. He, K. Jun, L. C. Gallington, H. Huo, C. J. Bartel, B. Ouyang, and G. Ceder, Selective formation of metastable polymorphs in solid-state synthesis. *Science Advances* 10, eadj5431 (2024).

21. L. Rettenberger, **N. J. Szymanski**, Y. Zeng, J. Scheutzke, S. Wang, G. Ceder, and M. Reischl, Uncertainty-aware particle segmentation for electron microscopy at varied length scales. *npj Computational Materials* 10, 124 (2024).

22. **N. J. Szymanski***, B. Rendy*, Y. Fei*, R. E. Kumar*, T. He, D. Milsted, M. J. McDermott, M. Gallant, E. D. Cubuk, A. Merchant, H. Kim, A. Jain, C. J. Bartel, K. Persson, Y. Zeng, and G. Ceder, An autonomous laboratory for the accelerated synthesis of novel materials. *Nature* 624, 86–91 (2023).

23. **N. J. Szymanski***, P. Nevatia*, C. J. Bartel, Y. Zeng, and G. Ceder, Autonomous and dynamic precursor selection for solid-state materials synthesis. *Nature Communications* 14, 6956 (2023).

24. **N. J. Szymanski**, C. J. Bartel, Y. Zeng, M. Diallo, H. Kim, and G. Ceder, Adaptively driven X-ray diffraction guided by machine learning for autonomous phase identification. *npj Computational Materials* 9, 31 (2023).

25. **N. J. Szymanski**, Z. Lun, J. Liu, E. C. Self, C. J. Bartel, J. Nanda, B. Ouyang, and G. Ceder, Modeling Short-Range Order in Disordered Rocksalt Cathodes by Pair Distribution Function Analysis. *Chemistry of Materials* 35, 4922–4934 (2023).

26. Y. W. Byeon, M. J. Gong, Z. Cai, Y. Sun, **N. J. Szymanski**, J. Bai, D. H. Seo, and H. Kim, Effects of cation and anion substitution in $KVPO_4F$ for K-ion batteries. *Energy Storage Materials* 57, 81–91 (2023).

27. S. Patil, D. Darbar, E. C. Self, T. Malkowski, V. C. Wu, R. Giovine, **N. J. Szymanski**, R. D. McAuliffe, B. Jiang, J. K. Keum, K. P. Koirala, B. Ouyang, K. Page, C. Wang, G. Ceder, R. J. Clément, and J. Nanda, Alternate Synthesis Method for High-Performance Manganese Rich Cation Disordered Rocksalt Cathodes. *Advanced Energy Materials* 13, 2203207 (2023).

28. **N. J. Szymanski**, Y. Zeng, T. Bennett, S. Patil, J. K. Keum, E. C. Self, J. Bai, Z. Cai, R. Giovine, B. Ouyang, F. Wang, C. J. Bartel, R. J. Clément, W. Tong, J. Nanda, and G. Ceder, Understanding the Fluorination of Disordered Rocksalt Cathodes through Rational Exploration of Synthesis Pathways. *Chemistry of Materials* 34, 7015–7028 (2022).

29. I. Khatri, **N. J. Szymanski**, B. B. Dumre, J. G. Amar, D. Gall, and S. V. Khare, Correlating structure and orbital occupation with the stability and mechanical properties of 3d transition metal carbides. *Journal of Alloys and Compounds* 891, 161866 (2022).

30. **N. J. Szymanski**, Y. Zeng, H. Huo, C. J. Bartel, H. Kim, and G. Ceder, Toward autonomous design and synthesis of novel inorganic materials. *Materials Horizons* 8, 2169–2198 (2021).

31. **N. J. Szymanski**, C. J. Bartel, Y. Zeng, Q. Tu, and G. Ceder, Probabilistic deep learning approach to automate the interpretation of multi-phase diffraction spectra. *Chemistry of Materials* 33, 4204–4215 (2021).

32. I. S. Khare, **N. J. Szymanski**, D. Gall, and R. E. Irving, Electronic, optical, and thermoelectric properties of sodium pnictogen chalcogenides: A first principles study. *Computational Materials Science* 183, 109818 (2020).

33. **N. J. Szymanski***, L. N. Walters*, D. Puggioni, and J. M. Rondinelli, Design of heteroanionic MoON exhibiting a Peierls metal-insulator transition. *Physical Review Letters* 123, 236402 (2019).

34. **N. J. Szymanski**, V. Adhikari, M. A. Willard, P. Sarin, D. Gall, and S. V. Khare, Prediction of improved magnetization and stability in $Fe_{16}N_2$ through alloying. *Journal of Applied Physics* 126, 093903 (2019).

35. **N. J. Szymanski**, I. Khatri, J. G. Amar, D. Gall, and S. V. Khare, Unconventional superconductivity in 3d rocksalt transition metal carbides. *Journal of Materials Chemistry C* 7, 12619 (2019).

36. B. B. Dumre, **N. J. Szymanski**, V. Adhikari, I. Khatri, D. Gall, and S. V. Khare, Improved optoelectronic properties in $CdSe_xTe_{1-x}$ through controlled composition and short-range order. *Solar Energy* 194, 742–750 (2019).

37. V. Adhikari, **N. J. Szymanski**, I. Khatri, D. Gall, and S. V. Khare, First principles investigation into the phase stability and enhanced hardness of TiN-ScN and TiN-YN alloys. *Thin Solid Films* 688, 137284 (2019).

38. **N. J. Szymanski**, L. N. Walters, O. Hellman, D. Gall, and S. V. Khare, Dynamical stabilization in delafossite nitrides for solar energy conversion. *Journal of Materials Chemistry A* 6, 20852 (2018).

39. **N. J. Szymanski**, Z. T. Y. Liu, T. Alderson, N. J. Podraza, P. Sarin, and S. V. Khare, Electronic and optical properties of vanadium oxides from first principles. *Computational Materials Science* 146, 310 (2018).

40. V. Adhikari, Z. T. Y. Liu, **N. J. Szymanski**, I. Khatri, D. Gall, P. Sarin, and S. V. Khare, First-principles study of mechanical and magnetic properties of transition metal (M) nitrides in the cubic $M_4N$ structure. *Journal of Physics and Chemistry of Solids* 120C, 197 (2018).

## Patents

2024 **Li extraction from spodumene**, A method of synthesizing Li precursors directly from hard minerals.

## Awards & Fellowships

2025 **NVIDIA Academic Grant Program Award**, NVIDIA

2024 **Didier de Fontaine Award in Theory and Computation** , UC Berkeley

2023 **Jane Lewis Fellowship for Research in Mining**, UC Berkeley
**Vedensky Award in Process Metallurgy**, UC Berkeley

2019 **NSF Graduate Research Fellowship**, National Science Foundation

## Software

**GALAXI.** A deep-learning framework for automated phase identification from powder XRD patterns.

**CrysTopo.** Tools for the topological analysis of DFT-computed charge densities.

**MatGen-Baselines.** A package to benchmark generative AI for materials discovery.

**XRD-AutoAnalyzer.** A neural network-based approach to XRD analysis.

**ARROWS.** A decision-making algorithm designed to optimize solid-state synthesis procedures.

**AdaptiveXRD.** Software that integrates automated XRD analysis with in-line steering of measurements.

## Teaching Experience

2026 **COM SCI M148: Introduction to Data Science**, Instructor
2026 **How Machine Learning Can Assist in Powder XRD**, Tutorial Instructor, MRS Spring Meeting
2026 **MAT SCI 104: Introduction to Engineering Materials**, Instructor
2026 **MAT SCI 90L: Physical Measurements of Materials**, Instructor
2024 **Machine learning for Chemical Engineering and Materials Science**, Guest Lecturer
2023 **Thermodynamics and Phase Transformations**, Graduate Teaching Assistant